

LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written examination (resit)

The use of electronic devices and additional material (books, notes, etc.) is not permitted during the exam. Justify clearly your answers and please write your anonymous code at the top of each sheet.

1. Determine the infinite simple continued fraction representation of the number $\sqrt{13}$.
2. (a) Show that 2 is a primitive root modulo 19.
   (b) Calculate a table of indices with respect to the primitive root 2 (modulo 19).
   (c) Find all solutions to the congruence
   $$6x^{15} \equiv 13 \pmod{19}.$$
3. For a positive integer n, let $\tau(n)$ be the number of divisors and $\omega(n)$ the number of distinct prime factors of n. Prove that
   $$\tau(n^2) = \sum_{d|n} 2^{\omega(d)}.$$
4. Evaluate the Legendre symbols $\left(\frac{71}{73}\right)$ and $\left(\frac{93}{191}\right)$ (the numbers 71, 73 and 191 are prime, but $93 = 3 \cdot 31$).
5. Let p be an odd prime such that
   $$p \mid a^2 + b^2$$
   for some positive integers a, b with $\gcd(a, b) = 1$. Prove that $p \equiv 1 \pmod{4}$.
6. Let $p = 4n + 1$ be a prime number, and let d be a divisor of n. Prove that $\left(\frac{d}{p}\right) = 1$.